

Publication 1

Title	A Machine Learning Framework towards Bank Telemarketing
Harvard reference	Tékouabou, S.C.K., Gherghina, Ş.C., Toulmi, H., Neves Mata, P., Mata, M.N. and Martins, J.M. (2022) ‘A machine learning framework towards bank telemarketing prediction’, <i>Journal of Risk and Financial Management</i> , 15(6), p. 269. doi:10.3390/jrfm15060269. MDPI
Link	Open publication
Modelling	The study developed a Class Membership-Based (CMB) classifier to predict whether a banking customer would respond positively to an offer. The authors compared the approach with previous machine-learning approaches and evaluated performance using accuracy, AUC and F1-score . Their CMB approach reached 97.3% accuracy and 95.9% AUC in their reported experiments. The model's decision function was designed to process nominal/categorical variables directly rather than relying entirely
Preprocessing techniques	Data were separated according to numeric, scaled/ordinal, Boolean and nominal variable types. Scaled attributes were converted to ordinal numbers; Boolean variables were converted to 0/1; nominal attributes were handled directly by the CMB classifier. Missing numeric/scaled values were replaced using the mean and missing Boolean/nominal values using the mode. The study also performed feature selection and standardisation/normalisation , retaining 18 significant variables in its
Datasets used	A Portuguese bank direct-marketing dataset , from the UCI Bank Marketing data family. The study describes about 45,000 banking customers contacted by telephone and offered a long-term deposit product. Fewer than 12% were in the positive-response class, making the problem
Key takeaways	1. STADIOEquities should treat activation as a binary classification problem , similar to predicting whether a banking customer will respond positively to a financial-product offer. 2. Categorical variables such as acquisition channel, onboarding status, region and investment goal must be encoded carefully because preprocessing decisions can materially affect performance. 3. Class imbalance must be considered because the customers who activate or fail to activate may not occur in equal

Publication 2

Title	Propension to Customer Churn in a Financial Institution: A Machine Learning Approach
Harvard reference	Lemos, R.A. de L., Silva, T.C. and Tabak, B.M. (2022) ‘Propension to customer churn in a financial institution: a machine learning approach’, <i>Neural Computing and Applications</i> , 34(14), pp. 11751–11768.
Link	Open publication
Modelling	Six supervised classifiers were compared: Decision Tree, K-Nearest Neighbours, Logistic Regression, Elastic Net, Support Vector Machine with radial kernel, and Random Forest . The researchers also constructed an ensemble of the six models. Model selection used repeated 10-fold cross-validation , while 90% of the data were used for training/model selection and 10% were retained as a hold-out test set. AUC-ROC was used as the optimisation measure. Random Forest achieved the strongest individual performance, with test ROC around 0.9015; the ensemble achieved around 0.9018 but was not statistically superior to Random Forest. PubMed Central (PMC)
Preprocessing techniques	Customers with insufficient relationship history were filtered according to the study's eligibility rules. Less than 6% of observations contained at least one missing attribute; missing data were imputed using KNN imputation with k=3 . Near-zero-variance variables were removed and numeric variables were standardised . The final modelling dataset contained 28 predictors. PubMed Central (PMC)
Datasets used	A proprietary customer-level dataset from a large Brazilian financial institution , containing approximately 500,000 observations after preprocessing. It contained behavioural information about customers' relationship with the institution, financial products and banking activity. The study defined churn using customer inactivity and investigated the prediction of churn over the following six months. PubMed Central (PMC)
Key takeaways	1. Compare several algorithms using the same train/test and cross-validation procedure rather than selecting one algorithm in advance. 2. Random Forest is an important candidate for the STADIOEquities dormancy model because it performed strongly on complex banking behaviour data. 3. Behavioural relationship variables—including frequency of financial-service usage and number of products held—were more predictive than simply examining account balances, suggesting STADIOEquities should make strong use of app activity, trades, deposits and product interactions rather than relying mainly on demographics.

Publication 3

Title	Investigating Customer Churn in Banking: A Machine Learning Approach and Visualization App for Data Science and Management
Harvard reference	Singh, P.P., Anik, F.I., Senapati, R., Sinha, A., Sakib, N. and Hossain, E. (2024) 'Investigating customer churn in banking: a machine learning approach and visualization app for data science and management', <i>Data Science and Management</i> , 7(1), pp. 7–16.
Link	Open publication
Modelling	Four classifiers were evaluated: Logistic Regression, Support Vector Machine, Random Forest and XGBoost . The authors created modelling pipelines that normalised the features and used grid-search cross-validation for model configuration. Performance was examined using accuracy, sensitivity, specificity, F1 and AUC. Models were evaluated both before and after addressing class imbalance. Boise State University
Preprocessing techniques	RowNumber , CustomerID and Surname were removed because they were not useful predictors. Categorical variables were converted to numerical form. Features were normalised to a common scale. Three engineered variables— TenureByAge , BalanceSalaryRatio and CreditScoreGivenAge —were created. Most importantly, SMOTE was applied to address the strong imbalance between churners and non-
Datasets used	A publicly accessible bank customer churn dataset from Kaggle containing 10,000 customers . Variables include credit score, geography, gender, age, tenure, balance, number of products, credit-card ownership, active-member status, estimated salary and the binary Exited target. Approximately 80% of observations represented non-churners and 20% represented churners. Boise State University
Key takeaways	1. Accuracy should not be used alone when predicting a minority class; sensitivity/recall, F1 and AUC must also be considered. Before SMOTE, some models showed good overall accuracy but poor sensitivity to actual churners. 2. Class balancing substantially improved the ability to identify churners, so SMOTE or an alternative should be tested for the STADIOEquities models if the target classes are imbalanced. 3. Feature engineering can convert basic account information into stronger behavioural indicators; similarly, STADIOEquities can create variables such as days-to-first-deposit, sessions-per-week, days since last trade, KYC

Publicly Available Dataset

Field	Dataset description
Name	Bank Marketing – bank-full.csv
Harvard reference	Moro, S., Rita, P. and Cortez, P. (2014) <i>Bank Marketing</i> [Dataset]. UCI Machine Learning Repository. doi:10.24432/C5K306. Dev Archive
Link	UCI Bank Marketing dataset
Number of instances	45,211 instances in bank-full.csv. UCI also supplies a smaller bank.csv sample containing approximately 10% of the observations and an additional version of the dataset with more socioeconomic variables. Dev Archive
Number of feature columns	16 predictor features , plus the binary target y , giving 17 columns in total in the bank-full.csv file. GitHub
Number of classes (if applicable)	2 classes: yes = the customer subscribed to the term deposit; no = the customer did not subscribe. Dev Archive
Target range (if applicable)	Not applicable because this is a classification problem , not regression.
Origin	The dataset contains records from direct-marketing campaigns conducted by a Portuguese banking institution . Customers were contacted by telephone, sometimes on multiple occasions, and the objective was to determine whether they ultimately subscribed to a term deposit. The data contain client characteristics, financial information and campaign/contact history. It is not described by UCI as synthetically generated ; it originates from historical banking marketing campaigns. Dev Archive